

Automotive Medallion Pipeline — Dashboard & Insights

Bronze → Silver → Gold lakehouse on Databricks · native AI/BI dashboard

All charts are computed on **petrol**, **showroom**, **light** vehicles within the valid horsepower range (50–2000 hp) — the Gold-layer data-quality filters.

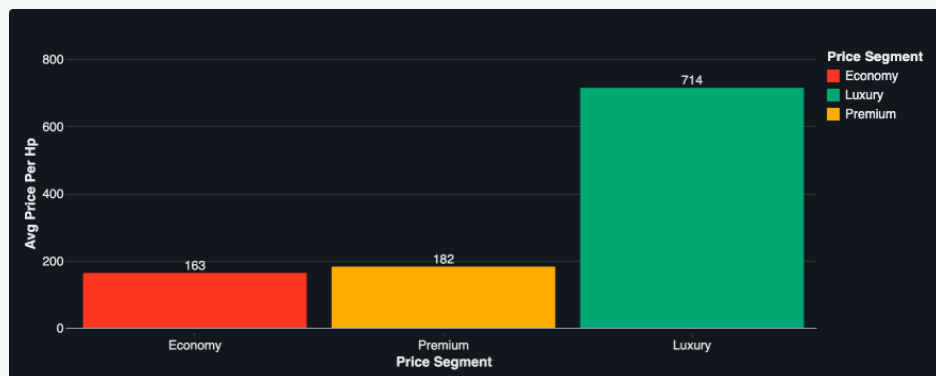
WHY THIS ANALYSIS

The source was **1,218 raw, free-text car listings** — prices, horsepower and fuel all recorded as messy strings. The pipeline cleans and structures that data, then answers one practical question: **how do price, power and value move across market segments, and where does a buyer's money actually go?**

IN SHORT

- Value per dollar collapses at the top: Economy and Premium cost ~\$163–182 per horsepower, but Luxury costs **\$714** — buyers there pay for brand and exclusivity, not extra output.
- Power scales cleanly with price (~159 → 336 → 634 hp per tier), so the luxury premium isn't explained by performance alone.
- The market is mostly mass-market — ~82% of petrol models are Economy/Premium — and these figures are trustworthy **only because impossible outliers** (a 2,488-hp van, a \$5M race prototype) were detected and excluded during cleaning.

Price efficiency — average price per horsepower, by segment



Why this analysis. Comparing raw prices across a \$20k hatchback and a \$2M hypercar says nothing about value. Normalising price by power — dollars per horsepower, petrol-only for a fair comparison — turns "expensive" into "expensive for what you get," the question a buyer actually cares about.

What it shows. Economy and Premium land close (\$163 vs \$182/hp), but Luxury leaps to **\$714/hp (~4×)**. Above a point the premium buys brand and exclusivity, not output.

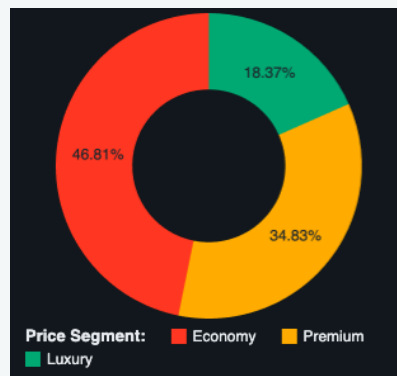
Power by segment — average horsepower



Why this analysis. The price-per-hp gap raises the obvious follow-up: is the luxury premium at least backed by more raw power? This isolates horsepower on its own, by segment, to separate "pay more and get more" from "pay more for the same."

What it shows. Power does climb steeply (~159 → 336 → 634 hp, roughly doubling per tier) — so luxury cars are more powerful, but price rises far faster than power, confirming the premium isn't really about performance.

Market composition — petrol models by segment



Why this analysis. Before reading anything into segment averages, we need to know where the cars actually sit — is this market (and dataset) driven by exotics or by mainstream models? Composition keeps the rare high-end tail from skewing the story.

What it shows. The ring is the share of petrol models per segment: ~82% are **Economy or Premium** (47% + 35%), Luxury a thin 18%. The typical car is mass-market; the exotics are a small, high-variance minority.

Built on Databricks (PySpark · Spark SQL · Delta Lake · Unity Catalog · AI/BI) — figures from the Gold tables `price_per_hp_petrol` and `power_sweet_spot`.